METRIC

DOD-STD-1476A
19 November 1986
SUPERSEDING
DOD-STD-1476
1 August 1977

MILITARY STANDARD

METRIC SYSTEM, APPLICATION IN NEW DESIGN

AMSC N/A AREA MISC

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
Washington, DC 20301

Metric System Application in New Design

DOD-STD-1476A

* 1. This Military Standard is approved for use where applicable by all Departments and Agencies of the Department of Defense.

* 2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST,



AMSC N/A AREA MISC

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.

FOREWARD

1. This standard is intended to be referenced only in those contracts for which a decision has been made to design an item, equipment, or system in metric units.

2. Care must be exercised in specifying metric components because if they are specified prior to industrial production to adopted national metric standards, it could result in: (a) nonstandard metric configurations, (b) unnecessary items in the supply system, and (c) extra costs because DoD contracts might absorb initial conversion costs of the producing industries.

3. It is usually more practical to design and produce a "new" item, equipment, or system in metric than to convert existing designs or production hardware. This standard requires: (a) that new designs for the end items under contract be designed and expressed in metric units, and (b) that existing metric designed components (parts and assemblies) be selected providing they are technically adequate and available at an equal or lower life-cycle cost.

4. During an extended period of changeover to the use of the metric system on a national basis, commercial and other already-designed components will be available in a mix of straight inch-pound, hybrid, and metric designs.

5. Tailoring of the use of this standard to meet the requirements of specific contract is encouraged.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph				Page
	*	1	SCOPE	1
		1.1	Scope	1
		1.2	Purpose	1
	*	2.	REFERENCED DOCUMENTS	1
		2.1	Government documents	1
		2.1.1	Standards	1
		2.2	Other publications	1
		2.3	Order of precedence	2
		3.	DEFINITIONS	2
	*	3.1	Metric, metric system, metric units	2
		3.2	Inch-pound	2
		3.3	Metric design	2
		3.4	Hard conversion	2
		3.5	Soft conversion	2
		3.6	Dual dimensions	2
		3.7	Dual indication	2
		3.8	Hybrid	3
	*	4.	GENERAL REQUIREMENTS	3
	*	5.	DETAILED REQUIREMENTS	3
		5.1	Metric units	3
		5.1.1	Preferred units	3
		5.2	New design	3
		5.3	Existing design	3
		5.3.1	Material sizes	3
		5.3.2	Components	3
		5.3.3	Instrumentation	4
		5.4	Modification of existing inch-pound designs	4
		5.5	Technical documentation	4
		5.5.1	Engineering drawings	4
		5.5.2	Specifications	4
		5.5.3	Other technical data	4
		5.5.4	Interfaces	4
		5.5.5	Inch-pound and metric equivalents	4
		5.5.6	Metric identifier	5
	*	6	NOTES	5
		6.1	Subject term (key word) listing	5
		6.2	Changes from previous issue	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.

1. SCOPE

* 1.1 Scope. This standard covers the general requirements for employing the metric system of measurement in new design and in accompanying documentation.

1.2 Purpose. The purpose of this standard is to describe how the metric system will apply to new design or existing subsystems/equipments, during design of new systems (ships, aircraft, missiles, etc.), in the selection of materials and components, and for configuration and functional description in accompanying documents.

2. REFERENCE DOCUMENTS

* 2.1 Government documents.

2.1.1 Standards. Unless otherwise specified, the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

* STANDARDS

Federal

FED-STD-376	Preferred Metric Units For General Use By The Federal Government

Military

MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of

(Copies of standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this standard to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. The issues of documents which have not been adopted shall be those in effect on the date of the cited DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 380 Metric Practice Guide

* (Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

(Nongovernment standards are generally available for reference from libraries. They are also distributed among nongovernment standards bodies and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

3. DEFINITIONS

As used herein, the expressions or terms are defined as follows:

3.1 Metric, metric system, metric units. The International System of Units (commonly abbreviated as SI), as established by the General Conference of Weights & Measures in 1960 and as interpreted or modified for the United States by the Secretary of Commerce.

3.2 Inch-pound units. The customary system formerly and currently used in the United States (foot, inch, pound, BTU, horsepower, degree Farenheit, etc.).

3.3 Metric design. Product design using metric dimensions, selected as appropriate, without considering conceptual or physical conversion from inch-pound units.

3.4 Hard conversion. The process of changing inch-pound measurement units to non-equivalent metric units which necessitates physical configuration changes outside those permitted by established measurement tolerances. NOTE: Although the term "hard conversion" is in general use, it is technically incorrect when applied to specific items because no "conversion" takes place. Instead, a new metric item requiring new item identification is created to replace the customary item. The new item is often referred to as being in "hard metric".

3.5 Soft conversion. The process of changing inch-pound measurement units to equivalent metric units within the acceptable measurement tolerances without changing the physical configuration. In other words, it is the same both before and after conversion.

3.6 Dual dimensions. The inclusion of linear dimensions in views of a drawing in both metric and inch-pound units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.

3.7 Dual indication. The inclusion, in text or on instrumentation and gaging, of a quantity (characteristic or dimension) in both metric and inch-pound units (for example: 700 kPa (100 lbf/in^2)).

3.8 Hybrid. A combination or mixture of metric and inch-pounds items.

4. GENERAL REQUIREMENTS

Not applicable.

5. DETAILED REQUIREMENTS

* 5.1 Metric units. Metric units, practices, and usages shall be in accordance with ASTM E 380.

5.1.1 Preferred units. Unless otherwise specified, the preferred metric units for the commonly-used quantities shall be in accordance with FED-STD-376.

5.2 New design. New items designed under contract shall be of metric design. Hybrid design, however, is acceptable where required to accommodate existing customary items selected in accordance with 5.3. Where applicable U.S. Government, non-Government (adopted for use by DoD), U.S. National or international documents exist which establish preferred metric modular sizes or other design standards, they shall be used to the maximum practical extent. Where a U.S. standard is established with greater definition or restriction than a prevailing international standard, the U.S. standard shall apply.

5.3 Existing items. Existing items shall be selected for use in accordance with the following guidelines, giving due consideration to interchangeability.

5.3.1 Material sizes. Metric sizes (e.g., sheet, plate, bar, rod, wire, etc.) shall be used when no cost or performance penalty will be incurred, or when additional first cost is justified by other considerations.

5.3.2 Components. Components shall be selected as follows:

a. Items designated as metric items in documents identified in Groups I, II, and III of MIL-STD-143.

b. All other items on the basis of lowest life cycle cost, whether metric or inch-pound.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.

5.3.3 Instrumentation. Unless otherwise specified, instrumentation and gaging (temperature, pressure, etc.) shall show the indications in metric units. The actual construction of the instrument or gage may be either metric or inch-pound in accordance with 5.3.2.

5.4 Modification of existing inch-pound designs. Unless otherwise specified, the extent to which metric design is required in the modification of existing inch-pound designs shall be determined on an individual case basis, considering technical and economical feasibility or other factors.

5.5 Technical documentation. Technical documentation shall comply with the following:

5.5.1 Engineering drawings.

a. New design. Values shall be expressed in metric units.

b. Existing design (including control drawings). Values shall be expressed in the unit system in which the item or items were designed.

c. Modified customary design. On new drawings prepared to describe new versions of existing inch-pound designs, values shall be expressed in the unit system applied to the modified portion of the design. When metric conversion of any part of the design is required, applicable values shall be expressed in metric units.

5.5.2 Specifications. Specifications prepared shall use the terminology of the unit system in which the item is to be designed.

5.5.3 Other technical data. Technical manuals, test reports, and other technical data shall use the terminology of the unit system in which the item is designed.

5.5.4 Interfaces. For features that interface between inch-pound and metric items, inch-pound and metric equivalents shall be specified directly on the drawing or in the document in accordance with 5.5.5.

* 5.5.5 Inch-pound and metric equivalents. Unless otherwise specified, the use of both inch-pound and metric equivalents is optional, except as required by 5.5.2, 5.5.3, and 5.5.4. When equivalents are included, they shall be specified as follows:

a. Dual indication (see 3.7). The metric value shall be stated first followed by the inch-pound value in parentheses.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.

b. Tabular form. Unless otherwise specified, table(s) may be included directly on the drawing or document. It shall translate all required values from one system of units to the other in ascending or descending order.

5.5.6 Metric identifier. A metric identifier, that is the word "METRIC", preferably enclosed in a rectangle, shall be placed on the field of the drawing near the title block. On other technical data, it shall be located in the vicinity of the document number. Lettering size shall be approximately the same as the drawing or document number. When nonmetric sheets are included in a metric document, the identifier shall be placed on each metric sheet only.

6. NOTES

* 6.1 Subject term (key word) listing.

Design, Metric
Hybrid
Metrication
System, Metric

* 6.2 Changes from previous issue. The margins of this standard are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evalute the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MI
Navy - SH
Air Force - 01

Review Activity:
Army - AV
Air Force: 70, 71, 80, 82, 84, 99
DLA - GS

User Activity:
Navy - YD
Air Force - 90

Preparing Activity:
Army - MI
(Project MISC-0029)

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Missile Command
ATTN: AMSMI-RD-SE-TD-ST
Redstone Arsenal, AL 35898-5270

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-STD-1476A	METRIC SYSTEM, APPLICATION IN NEW DESIGN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:29Z
Check the source to verify that this is the current version before use.